

DoS and DDoS

In this module we will focus Denial-of-Service(DoS) and Distributed Denial-of-Service(DDoS) attacks. This module cover understanding of different DoS and DDoS attacks, attacking techniques, concept, attacking tools and its countermeasures. How to defend against these attacks.

What is a Denial-of-Service Attack?

Denial of Service (DoS) is an attack on a computer or network that reduces, restricts or prevents accessibility of system resources to its legitimate users.

In a DoS attack, attackers flood a victim system with nonlegitimate service requests or traffic to overload its resources

DoS attack leads to unavailability of a particular website and slow network performance.

What are Distributed Denial of Service Attacks?

A distributed denial-of-service (DDoS) attack involves a multitude of compromised systems attacking a single target, thereby causing denial of service

for users of the targeted system.

To launch a DDoS attack, an attacker uses botnets and attacks a single system.

DoS and DDoS Impact

Loss of Goodwill

Disabled Network

Financial Loss

Disabled Organization

Like Flipkart got down in past.

DoS/DDoS Attack Techniques

Bandwidth Attacks and Service Request Floods

SYN Attack

SYN Flooding Attack



ICMP Flood Attack

Botnet Attack

And many more

Bandwidth Attacks

When a DDoS attack is launched, flooding a network, it can cause network equipment such as switches and routers to be overwhelmed due to the significant

statistical change in the network traffic.

Attackers use botnets and carry out DDoS attacks by flooding the network with ICMP ECHO packets.

Basically, all bandwidths is used and no bandwidth remains for legitimate use.

Service Request Floods

An attacker or group of zombies attempts to exhaust server resources by setting up and tearing down TCP connections.

Service request flood attacks flood servers with a high rate of connections from a valid source

It initiates a request on every connection

SYN Attack

The attacker sends a large number of SYN request to target server (victim) with fake source IP addresses.

The target machine sends back a SYN/ACK in response to the request and waits for the ACK to complete the session setup

The target machine does not get the response because the source address is fake.

SYN Flooding Attack

SYN Flooding takes advantage of a flaw in how most hosts implement the TCP three-way handshake.

When Host B receives the SYN request from A, it must keep track of the partially opened connection in a "listen queue" for at least 75 seconds. A malicious host can exploit the small size of the listen queue by sending multiple SYN requests to a host, but never replying to the SYN/ACK. The victim's listen queue is quickly filled up. The ability of holding up each incomplete connection for 75 seconds can be cumulatively used as a Denial of Service attack.

ICMP Flood Attack

ICMP flood attack is a type DoS attack in which attacker sends a large number of ICMP packets.

Practical Part :

1. Syn Flooding Attack Using Metasploit :

Commands -

```
#nmap -p 21 192.168.1.132
#msfconsole
#use auxiliary/dos/tcp/synflood
#show options
#set RHOST <victim ip address>
#set RPORT 21
#set SHOST <spoofable ip address>
#set TIMEOUT 30000
#exploit
```

Now open Victim Machine and Wireshark in our system to check the performance and packet flow.

2. Syn Flooding Attack Using Hping3 :

Hping3 - It is a network tool, which is used for crafting the packets, testing the firewall, digital footprinting and much more.

Use command `#hping3 <victim ip address> --flood`



Now open Victim Machine and Wireshark in our system to check the performance and packet flow.

Botnets :

A botnet is a huge network of the compromised systems and can be used by an attacker to launch denial-of-service attacks.

Ping of Death attack :

The attacker aims to disrupt a targeted machine by sending a packet larger than the maximum allowable size, causing the target machine to freeze or crash.

IPv4 ping packets are much larger, and can be as large as the maximum allowable packet size of 65,535 bytes. Some TCP/IP systems were never designed to handle packets larger than the maximum, making them vulnerable to packets above that size.

Ping of Death Attack Using Hping3 :

```
#hping3 192.168.195.183 -c 10000000000 -d 999999999 --rand-source --flood -p 21
```

- We can modify this command according to our need and we can use multiple

systems and multiple terminal for this attack.

Now open Victim Machine and Wireshark in our system to check the performance and packet flow, or

Check the site status after DDOS:- <https://isitdown.us/>

<https://www.isitdownrightnow.com/>

How to Prevent or defend against a DoS or DDoS Attack :

1. Traffic Analysis and Filtering
2. Blackhole or Sinkholing
3. IP-based Prevention
4. Shutting Down the services



1. Traffic Analysis and Filtering - In this we monitor the data and traffic flowing into our network. We try to filter this.

2. Blackhole or Sinkholing - In this prevention we try to redirect the waste traffic to some other place. Basically we put limit on our system for the coming packet, and

the extra packet will be redirected to other place.

For better understanding we have some videos :

<https://www.youtube.com/watch?v=mf6OMPnfLN8>

https://www.youtube.com/watch?v=yPNKQZar-Fw&ab_channel=HackerSploit

3. Ip-based Prevention - In this our IDS And IPS helps us in collecting the packet information and prevent the attack. Our IDS will detect the abnormal packets activity

and it will send logs to admin after that admin will block the Malicious Ip.

Until in some cases our IPS can protect the attack.

